

ABHIJITH SIVAPRASADAN

Thermal-Fluid / CFD / Test & Validation Engineer | Gas Turbine Test-Rig CFD/CHT | Python | NI-DAQ/LabVIEW
Solna, Stockholm, Sweden | Previously based in Norrköping during Siemens Energy thesis | Open to Finspång / Linköping /
Stockholm / hybrid assignments
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Portfolio: [abhijith-sivaprasadan.github.io](https://github.com/abhijith-sivaprasadan) | github.com/abhijith-sivaprasadan

SUMMARY

Mechanical and sustainable energy engineer with Siemens Energy Finspång thesis experience in high-temperature gas-turbine test-rig hardware, steady-state compressible CFD, conjugate heat transfer, pressure-loss analysis, thermal KPI evaluation, Python post-processing, NI-DAQ/LabVIEW measurement workflows and validation planning. Strong fit for Quest Global contractor assignments supporting Siemens Energy in thermal-fluid analysis, CFD/CHT, test and validation support, calculation workflows, mechanical design support and technical documentation.

TARGET QUEST / SIEMENS ENERGY ASSIGNMENTS

- CFD / CHT support: pressure-loss studies, thermal delivery analysis, mesh checks, sensitivity studies, post-processing and result interpretation.
- Test and validation support: NI-DAQ/LabVIEW measurement workflows, sensor context, measurement-chain readiness, test planning and troubleshooting documentation.
- Calculation / analysis engineering: Python-supported KPI extraction, plots, comparison tables, repeatable calculations and assumptions tracking.
- Mechanical design support: geometry preparation, CAD/model cleanup, drawing/documentation support and mechanical design review assistance.
- Technical documentation: validation notes, boundary-condition summaries, model limitations, internal reports and engineering handover material.

CORE SKILLS

CFD / thermal-fluid	ANSYS Fluent, CFD/CHT, compressible flow, heat transfer, pressure-loss analysis, k-omega SST, mesh/near-wall checks, sensitivity studies, validation planning.
Gas turbine context	Siemens Energy Fluid Dynamic Lab, high-temperature test-rig hardware, gas-turbine test environment, thermal delivery, pressure drop and validation constraints.
Testing / instrumentation	NI-DAQ, NI MAX, LabVIEW workflows, thermocouples, static/dynamic pressure sensors, measurement-chain checks, test readiness, troubleshooting documentation.
Programming / analysis	Python, pandas, NumPy, SciPy, matplotlib, MATLAB basics, Excel, KPI extraction, technical plotting, repeatable engineering calculations, Git.
CAD / design support	Siemens NX exposure, Teamcenter PLM2020 context, SpaceClaim, SolidWorks foundation, geometry cleanup, model preparation, technical documentation.
Target ramp-up areas	STAR-CCM+, Siemens in-house calculation workflows, Cantera/combustion analysis, deeper combustor and gas-turbine component workflows.

PROFESSIONAL EXPERIENCE

Test Engineer / Master Thesis Student - Fluid Dynamic Lab

Siemens Energy AB, Finspång, Sweden | Apr 2025 - Nov 2025

- Completed master thesis in an industrial gas-turbine test-rig R&D environment focused on high-temperature hardware.
- Developed steady-state compressible CFD and CHT models in ANSYS Fluent to evaluate pressure-loss and thermal behaviour.
- Worked with boundary-condition definition, model assumptions, material/thermal constraints and validation-oriented interpretation.
- Performed mesh/near-wall checks, sensitivity studies and KPI-based case comparison to support engineering interpretation.
- Used Python for post-processing, pressure-loss/thermal KPI extraction, technical plots and repeatable analysis workflows.
- Supported measurement-chain readiness using NI-DAQ, NI MAX and LabVIEW workflows with thermocouples and pressure sensors.
- Documented heater-related test observations and troubleshooting notes, including likely contributors and practical mitigation ideas.
- Prepared technical documentation covering assumptions, boundary conditions, limitations, validation planning and engineering conclusions.
- Built familiarity with Siemens Energy-style engineering communication, test-rig constraints, documentation discipline and cross-functional R&D work.

Energy Efficiency Intern

Alleima AB, Sandviken, Sweden / Remote | Dec 2024 - Jun 2025

- Developed an ISO 50001/EU EED-aligned energy performance mapping approach for industrial sites.
- Defined KPIs/EnPIs, load-driver normalisation logic, metering readiness and measurement planning.
- Worked with industrial utility themes including process heat, compressed air, heat recovery, data quality and improvement prioritisation.
- Prepared technical reports and improvement-roadmap logic linking energy analysis to operational actions.

PROFESSIONAL EXPERIENCE CONTINUED

Backend Engineer

QBurnst, Thiruvananthapuram, India | Aug 2021 - Jul 2023

- Developed backend APIs and services using Python/TypeScript-related workflows, Postman, Git and Docker.
- Worked with structured debugging, automated testing, version control, documentation and maintainable delivery in agile teams.
- Built transferable skills in reproducible workflows, traceability, issue diagnosis and technical communication.
- Experience is useful for engineering automation, CFD post-processing workflows, scripting, data handling and toolchain ramp-up.

SELECTED PROJECTS

High-Temperature Gas-Turbine Test-Rig CFD/CHT Case Study

Siemens Energy / KTH thesis portfolio case | 2025

- Documented industrial context, problem definition, CFD/CHT approach, boundary-condition logic, validation limits and engineering outcomes.
- Highlighted thermal-fluid reasoning, test-rig constraints, measurement planning and decision-ready reporting for engineering reviewers.
- Prepared a sanitized public-facing case study to communicate the method without exposing confidential Siemens Energy details.

Mechanical Design and SAE Mobility Projects

Bachelor-level engineering work | SolidWorks, drivetrain concepts, technical documentation | 2018 - 2021

- Created and supported CAD-based mechanical design work using SolidWorks for SAE Bicycle Design, BAJA and Tractor Design Competition activities.
- Contributed to drivetrain, transmission and component selection discussions and helped prepare technical/design documentation.
- Built practical understanding of manufacturability, assembly thinking, mechanical constraints and team-based engineering delivery.

District Heating Dispatch Optimisation

KTH course project | Python, PuLP/MILP | 2025

- Built multi-asset district-heating dispatch model including CHP, heat pumps, storage, outages and electricity-price coupling.
- Compared cost and emissions objectives, technology mix, seasonal operation and KPI outcomes.

ADDITIONAL TECHNICAL STRENGTHS

- Strong in linking analysis outputs to engineering decisions: KPI comparison, assumptions, limitations, risks and next-step validation needs.
- Comfortable working between simulation, data, test hardware context and documentation instead of treating analysis as a black-box activity.
- Able to ramp into new toolchains quickly due to prior software-engineering background and structured debugging experience.
- Interested in Siemens Energy / Quest tasks where quality, repeatability, documentation and collaboration with senior engineers matter.

EDUCATION

M.Sc. in Sustainable Energy Engineering

KTH Royal Institute of Technology, Stockholm, Sweden | 2023 - 2026

Relevant: Applied Heat and Power Technology, Numerical Heat Transfer, Sustainable Power Generation, Energy Systems Modelling, Practical Optimization of Energy Networks, Energy Management.

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India | 2017 - 2021

Relevant: thermodynamics, fluid mechanics, heat and mass transfer, thermal engineering, machine design, manufacturing, CAD/CAE and mechanical project documentation.

CERTIFICATIONS

Applied Computational Fluid Dynamics, Siemens; SSG Entre / Industrial Safety, Sweden; Schneider Electric - Assessing a Commercial Product's Sustainability; IBM Python for Data Science; Introduction to Git and GitHub; Digital Manufacturing and Design Technology.

AVAILABILITY

Currently based in Solna, Stockholm. Previously based in Norrköping during the Siemens Energy thesis period. Open to Finspång / Linköping / Stockholm-region assignments, including Quest Global contractor opportunities supporting Siemens Energy in CFD, thermal-fluid analysis, gas-turbine testing, validation, calculation engineering, mechanical design support, technical documentation or engineering analysis.